

Meaningful Use to Secure Interoperability

Addressing Cybersecurity Risks in Healthcare

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HHH 503 - Healthcare Cybersecurity: Regulations, Confidentiality, and Privacy

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August 17, 2026

The Health Information Technology for Economic and Clinical Health (HITECH) Act accelerated the adoption of electronic health records (EHRs), while the Meaningful Use program provided financial incentives beginning in 2011 to expand EHR adoption by encouraging the use of technologies such as clinical decision support (CDS), computerized provider order entry (CPOE), electronic prescribing (e-prescribing), and health information exchange (HIE), which improved access to health information through interoperability and contributed to improvements in quality of care (Centers for Medicare & Medicaid Services [CMS], 2023; Kan, 2011; Jones et al., 2014). Interoperability is the ability of different systems to communicate and exchange information across networks and domains. This paper argues that its security consequences vary by technology and therefore require multiple security controls to protect patient information and support secure interoperability. Digital Imaging and Communications in Medicine (DICOM) creates significant integrity and availability concerns in interconnected imaging environments, while FHIR introduces distinct authorization and confidentiality risks through application programming interface (API)-based data exchange.

As healthcare organizations expanded their exposure footprint through interoperability and exchanged information across access points, the number of dependencies requiring protection also increased (Office of the National Coordinator for Health Information Technology [ONC], 2026; Coventry & Branley, 2018). The acceleration of technology through HITECH and Meaningful Use produced new cybersecurity responsibilities associated with interoperability.

Fast Healthcare Interoperability Resources (FHIR) shows how increased interoperability can create new cybersecurity challenges by enabling application programming interface (API)-based communication among EHR platforms, mobile applications, patient portals, and other healthcare technologies (Health Level Seven International [HL7], 2023; Saripalle et al., 2019).

As interoperability expands the number of systems and applications that can access and exchange sensitive health information, it also increases opportunities for unauthorized access and cyberattacks that can lead to data breaches. The Common Vulnerabilities and Exposures (CVE) system provides standardized identifiers for tracking cybersecurity vulnerabilities in technology (Board of Cyber, n.d.). For example, CVE-2026-24487 identified a vulnerability in the OpenEMR FHIR CareTeam endpoint that allowed patient-scoped tokens to access care team data belonging to other patients (National Institute of Standards and Technology [NIST], 2026; SentinelOne, 2026). This vulnerability highlights how authentication alone is not enough; organizations must also verify that each endpoint API request is authorized for the specific patient information requested. Encryption mechanisms such as Transport Layer Security (TLS) can protect FHIR data during transmission, but they do not determine whether an authenticated user is authorized to access another patient's information. Because FHIR enables the exchange of clinical information across interconnected systems, authorization failures can become patient-safety concerns when inappropriate information enters clinical treatment workflows. Securing interoperability requires reliable information exchange and appropriate patient-level access controls.

Digital Imaging and Communications in Medicine (DICOM) is another example of how interoperability can introduce cybersecurity risks within healthcare systems (National Electrical Manufacturers Association [NEMA], n.d.; Desjardins et al., 2020). DICOM-enabled imaging systems have improved teleradiology by enabling radiologists to interpret images remotely and access results more efficiently (CARE, 2025). However, Black Hat research identified 18.2 million patient records in the United States that were exposed to the internet through insecure DICOM servers. This exposure can increase the risk of ransomware attacks, data breaches,

identity theft, and fraud (Yazdanmehr, 2023a, 2023b). These risks can lead to broader cybersecurity concerns in healthcare, where interconnected systems and vulnerabilities can lead to data breaches and operational disruption (Coventry & Branley, 2018; Kruse et al., 2017). Internet-exposed DICOM servers can allow attackers to modify images or introduce malicious code to connected picture archiving systems (PACS) and other interconnected systems (Desjardins et al., 2020). This risk is particularly significant because the integrity and availability of imaging data directly affect clinical decision-making. Altered images could undermine diagnostic accuracy, while ransomware or other disruptions could prevent clinicians from accessing necessary images, delay treatment, and result in patient harm or death (Computer Fraud & Security, 2020). Insecure legacy imaging systems can further increase these risks when they remain connected to newer technologies without adequate security controls. DICOM therefore demonstrates why interoperability security must protect not only patient information but also the integrity and availability of clinical systems.

The FHIR and DICOM examples confirm that interoperability requires layered security controls matched to the risks created by each technology. For interoperable systems, zero trust principles, multi-factor authentication (MFA), and role-based access control (RBAC) can strengthen identity and authorization. RBAC also supports the HIPAA Privacy Rule's minimum necessary standard by limiting access to information required for a user's intended function (U.S. Department of Health and Human Services [HHS], n.d.-a). TLS 1.3 protects data in transit by converting readable information into ciphertext that requires appropriate cryptographic keys to decrypt. Encryption of stored data provides additional confidentiality, although storage encryption can create data-sharing and access-design trade-offs. Network segmentation can limit the spread of a compromise across interconnected systems by isolating systems from reaching

sensitive areas of the network. Regular patching and maintenance can address known vulnerabilities identified through CVE tracking and reduce the risk of system outages (Vukotich, 2023). MFA, RBAC, patching, and zero trust are particularly important for FHIR because they address authentication, authorization, and vulnerabilities that can affect API-based data exchange, while segmentation is important for DICOM because it can contain compromises across interconnected systems and limit the impact of data breaches on clinical operations. Together, these controls address unauthorized access, data manipulation, and system disruption while supporting patient safety.

In conclusion, Meaningful Use improved access to health information and interoperability, but this acceleration in growth created more complex security responsibilities. Cybersecurity must mature in tandem with interoperability while addressing both legacy and emerging technical debt. Although the HIPAA Security Rule requires administrative, physical, and technical safeguards to protect ePHI, it does not prescribe specific technologies or a particular cybersecurity framework for compliance (HHS, n.d.-b). Healthcare organizations must apply risk-based practices such as system patching, legacy modernization, risk management, and vendor oversight to achieve secure interoperability beyond basic HIPAA compliance requirements. Future interoperability initiatives should evaluate security maturity alongside connectivity rather than treating cybersecurity as a secondary consideration. Ultimately, interoperability should be measured and managed by considering the trade-offs between information accessibility and cybersecurity restrictions and evaluating whether the systems that deliver information can be trusted when patient care depends on them.

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ChatGPT was used to assist with grammar, spelling, punctuation, clarity, and verification of information and citations. The author reviewed and evaluated all suggestions and is responsible for the accuracy, content, and final submission of this paper.